

16-bit Security Controller – V02 with PG-USON-10-2

Package Qualification Report

About this document

Scope and purpose

This document provides information about the results of the qualification of delivery form(s).

Intended audience

This Qualification Report is intended for customers of Infineon Technologies and certification parties. The data contained in this document is exclusively intended for technically trained staff.

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General Information

1 General Information

Table 1 General Information

Wafer Process Technology	C65FL
Wafer Production Site(s)	TSMC Fab14, Tainan/Taiwan
Pre-Assembly Site	Infineon Regensburg, Germany
Assembly Site	AMKOR (ATP), Philippines
Product Specification	Actual Version available at Customer Service Center
Errata Sheet	Actual Version available at Customer Service Center
PCN / Inof Note	---

1.1 Applied Qualification Standards

- JESD47
- ISO 7816
- ISO 10373
- ISO 14443

Family Qualification with Generic Data from Reference Products

2 Family Qualification with Generic Data from Reference Products

Table 2 Family Qualification

	Product	Wafer Fab	Wafer Technology	Assembly Technology	Assembly Technology
Part to be Qualified	16-bit Security Controller – V02	TSMC	C65FL	ATP	PG-USON-10-2
Reference Part A	M5072G	TSMC	L90FL	ATP	PG-VQFN-32-13
Reference Part B	M7892A	Infineon Dresden	L90FL	ATP	PG-USON-10-1
Reference Part C	---	---	---	---	---

Reference part types / products utilize the same fabrication process, design rules, and similar circuits.

Qualification Result Overview

3 Qualification Result Overview

Table 3 Qualification Result Overview

Qualification Test	Test Conditions	Part to be Qualified			Ref. Part A	Ref. Part B
		Lot #1	Lot #2	Lot #3		
Preconditioning (MSL1)	30 °C, 60 % r.h., 192 h soak, 260 °C, 3 x reflow JESD22-A113	0 / 231	0 / 231	0 / 231	-	-
High Temperature Storage Life	T _A =150 °C, 1000 h, JESD22-A103	0 / 77	0 / 77	0 / 77	-	-
Temperature Humidity Bias Life Test ¹	85 °C, 85 % r.h., 49 kPa, V _{DD} =5.5 V, 1000 h JESD22-A101	0 / 77	0 / 77	0 / 77	-	-
Temperature Cycling ¹	-55 °C...+125°C, 2 cycles/ hour, 700 cycles JESD22-A104	0 / 77	0 / 77	0 / 77	-	-
Unbiased HAST ¹	130 °C / 230 kPa, 85 % r.h., 96 h JESD22-A118	0 / 77	0 / 77	0 / 77	-	-
Electrostatic Discharge (CDM)	500 V JESD22-C101	0 / 3	-	-	-	-
Wire Bond Pull Strength	5 bonds / unit MIL-STD-883 Method 2011	0 / 8	0 / 8	0 / 8	-	-
Wire Bond Shear Strength	5 bonds / unit JESD22-B116	0 / 8	0 / 8	0 / 8	-	-
Solderability	Aging, wetting, de-wetting JESD22-B102	-	-	-	0 / 135	-
Package Dimensions	JESD22-B100	-	-	-	-	0/30

¹ After preconditioning

4 Qualification of μ Controller and References

For qualification of the μ Controller please refer to the corresponding qualification reports.

These documents are available at our Customer Service Center.

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Revision History

Revision History

Major changes since the last revision

Page or Reference	Description of change
Revision 1.0, 2018-01-31	
All	First issue

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Document reference

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